Commercial Portfolio Analysis

Assurance Bank, Alberta - Canada

Stephen Dugbartey
Edmonton, Alberta – Canada
+1 (825) 777-0501
ddstephen07@gmail.com

March 2026

1.0 Executive Summary

The commercial loan portfolio book is critical in driving income and value to the operation of the bank into the foreseeable future. An imaginary bank - **Assurance Bank** was used in this project.

Problem Statement:
The bank could not claim possession of the collateral of of a recently defaulted customer due to imperfect collateral registration into the bank's custody.

Project Objective:
The objective of this project analyzes the bank's commercial loan portfolio quality, risk and growth factors. The key determinants are:

- Credit worthiness of customer entry into the portfolio
- Loan processing TAT analysis
- Portfolio industry distribution
- Overdraft utilization and portfolio growth
- Interest income generation
- Non-performance risk tracking
- Collateral management (perfection and monitoring)

Project Scope:

- The commercial overdraft portfolio is the focus
- All customers operate within the province of Alberta
- 7300 commercial overdraft customers make up the portfolio since 2009 and 2026
- 797 new commercial overdrafts was approved in Q1 2026
- The portfolio was spread among 8 industry sectors
- Overdraft interest ranges from 11% to 21% based on the industry requirements
- All newly onboarded customers in Q1 2026 have 12 months overdraft tenor from the date of approval
- Approved TAT for loan processing is 15 days
- Credit score breakdown used in analysis is as follows:
  760 and above: Excellent
  729 - 759: Very Good
  660 - 728: Good
  560 - 659: Fair
  559 and below: Poor

Tools Deployed:

- Power query
- Power BI

Summary Project Findings:
The commercial overdraft portfolio is well balanced with income generation, utilization, growth and diversity across the 8 industries of lending but heavily strong on performance and collateral monitoring. It shows 0% overdue overdraft and 100% collateral monitoring. However, it contains 5.21% poorly credit-rated customers, 36% delayed processing and 0.6% collateral not perfected with exposed values at risks of $53.84m, $314.39m and $5.91m respectively. A risk metrics determined the impact of the 3 weaknesses identified. The poorly credit-rated customers and delayed processing times are HIGH risk and the collateral not perfected are MEDIUM risk. Strategies have been recommended to reduce these risk levels to an estimated LOW.

2.0 Methodology

The following methods were deployed:

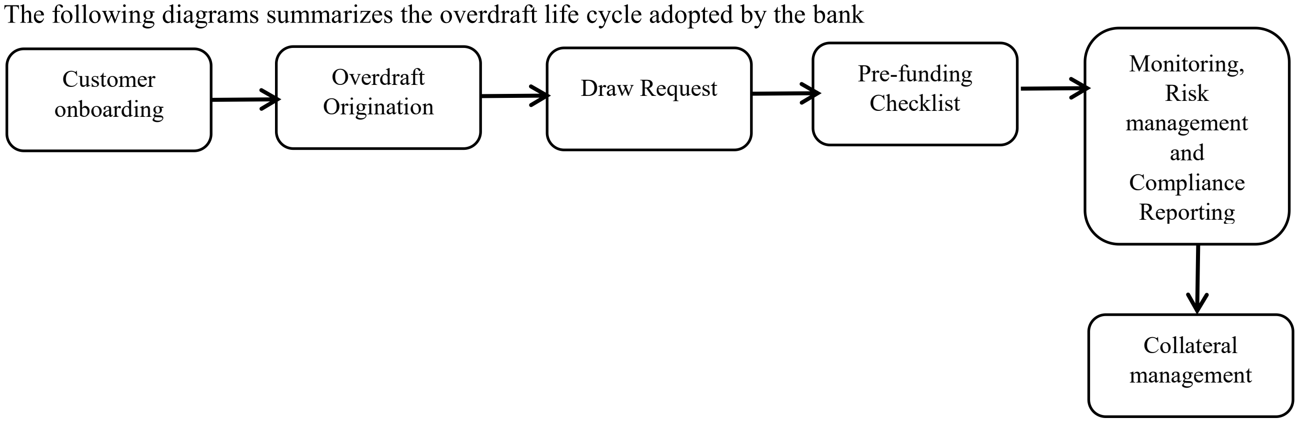
- 4 tables were created under the following names: ***customer_list, overdraft_origination, overdraft_utilization, overdraft_income***
- The various key indicators defined in the objective statement were analyzed using Power Query
- Interactive dashboards summary was created using Power BI
- Risk analysis was performed to determine the likelihood and impact of the weaknesses that exist in the portfolio
- Recommendations to be considered were suggested to aid resolve weaknesses discovered after the data analysis.

Data Limitations

- **Value Allocation Assumptions:** Overdraft amounts, interest income, utilization rate were estimated rather than direct attribution, affecting precision.
- **Limited Time Horizon:** Analysis based on a single period (FY2026) may not capture seasonality or long-term trends over a range of years
- **Exclusion of External Factors**: Market conditions, competitors' pricing and macroeconomics impacts were not fully incorporated.

Overdraft Lifecycle

The following diagrams summarizes the overdraft life cycle adopted by the bank



2.0 Project Outcomes

Based on the project objective, the following key performance areas were discovered:

Strengths

- The number of customers and overdraft exposure are fairly distributed across the 8 industries without concentration risk
- Average overdraft utilization ranges between 1% and 70% over the first quarter of 2026.
- The overdraft earned incomes of $3m, 7m and 23m over the first quarter of 2026
- The portfolio experiences 0% overdue balances
- 100 of the collateral have been monitored within Q1 2026

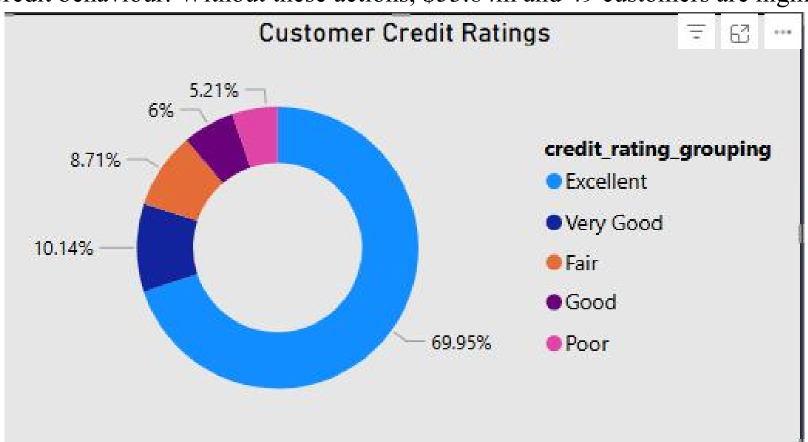
Weaknesses

- 5.21% of customers rated poor on credit score
- 36% of the overdrafts were approved beyond the 15 day processing TAT
- 0.6% of the collateral are yet to be fully and legally perfected

Details of Key Performance Areas Identified

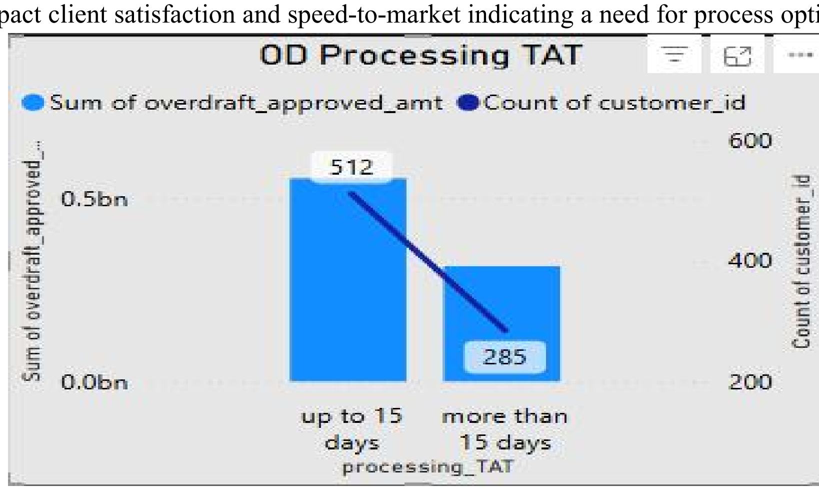
Credit worthiness of customer entry into the portfolio

5.21% of customers rated poor on credit score. Even though this segment is small, it presents higher risk clients within the portfolio, presenting an opportunity for targeted monitoring and risk-based pricing to protect returns. Additionally, evidence of exceptional approval is required from authorized persons with justification supporting reasons for granting credit to customer who perform poorly with credit behaviour. Without these actions, $53.84m and 49 customers are highly are risk.



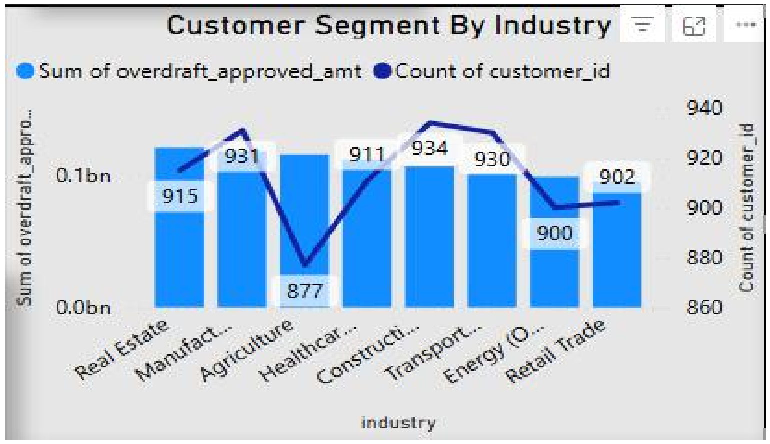
Loan processing TAT analysis

36% of the overdraft were approved beyond the 15 day processing TAT. It represents delays and operational inefficiencies in credit processing may impact client satisfaction and speed-to-market indicating a need for process optimization.



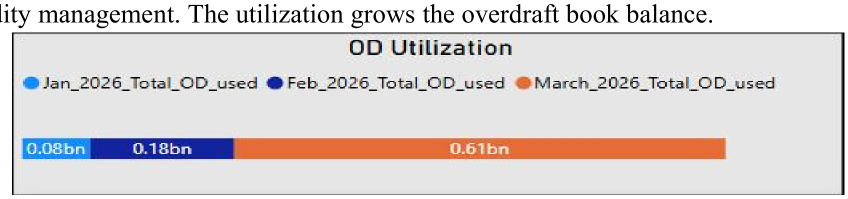
Portfolio industry distribution

The credit portfolio is evenly distributed across 8 industries. It is well-diversified, reducing sector-specific shocks and enhancing overall portfolio stability.



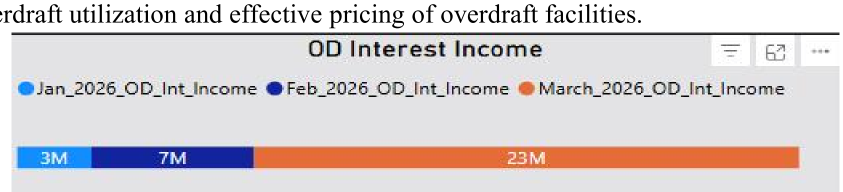
Overdraft utilization and portfolio growth

Average overdraft utilization ranges between 1% and 70% over the first quarter of 2026. It shows that clients are actively using their facilities indicating income generation for the bank and healthy engagement of customers while maintaining room for future lending and liquidity management. The utilization grows the overdraft book balance.



Interest income generation

The overdraft earned incomes of $3m, 7m and 23m over three months of Q1 2026. This shows a strong income growth over Q1 suggesting increased overdraft utilization and effective pricing of overdraft facilities.

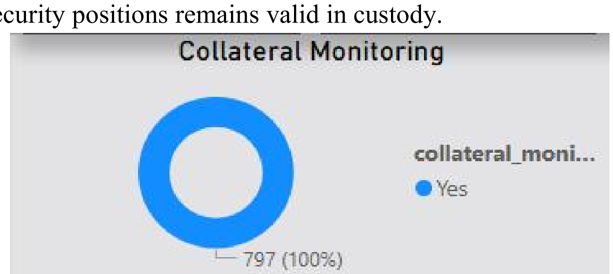


Non-performance risk tracking

The portfolio experiences 0% defaults showing strong quality, reflecting prudent underwriting and effective risk monitoring processes.

Collateral management (perfection and monitoring)

There exist a minor gap in legal perfection of collateral. Even though minimal, it exposes the bank to legal and recovery risk, requiring immediate remediation to ensure enforceability of security. Additionally, the bank has maintained strong collateral oversight, ensuring risk is continuously mitigated and security positions remains valid in custody.



3.0 Risk Metrics

The risk metrics is defined by the risk level rating, likelihood rating and impact rating.

Risk Level

Risk level rating is defined by the **likelihood rating** and **impact rating** of each risk area.

Risk Level	Rating
Low Risk	2 and below
Medium Risk	3 - 4
High Risk	5 and above

Likelihood

Defined as the possibility of the risk area occurring. The number of the affected overdraft expressed as a percentage of the total number of overdraft approvals defines the likelihood.

Likelihood	Percentage Referred	Rating
Unlikely	Less than 5%	1
Likely	Between 6% and 40%	2
Very Likely	Higher than 40%	3

Impact

Defined as the overdraft amount value that can be affected in relation to the expressed likelihood.

Impact Severity	Financial Impacted	Rating	Investigation and Resolution Timeline
Minor	Less than $5,000,000	1	Up to 2 months
Serious	Between $5,000,001 and $8,000,000	2	Up to 1 month
Major	Higher than $8,000,000	3	Up to 1 week

Commercial Portfolio Risk Assessment Summary

The three weaknesses identified have been assessed in the table below:

Risk Category	Likelihood		Financial Impact Severity		Risk Rating (likelihood rating + financial impact rating)	Risk Level
	Percentage	Rating	Amount	Rating		
5.21% of customer rated poor on credit score	5.21%	2	$53.84m	3	5	High
36% of the overdraft were approved beyond the 15 day processing TAT	36%	2	$314.39m	3	5	High
0.6% of the collateral are yet to be legally perfected	0.6%	1	$5.91m	2	3	Medium

4.0 Strategy Recommendations

The following strategies have been recommended to aid resolve the identified weaknesses with the commercial portfolio of Assurance Bank in Alberta.

Identified Weaknesses	Recommendation
5.21% of customer rated poor on credit score	● Adjust interest rates and exposure limits for lower-rated customers to compensate for higher risk and protect profitability. ● Track financial performance, utilization, and repayment behavior more frequently to detect early signs of deterioration. ● Enforce stricter covenants and require more frequent financial reporting to maintain tighter control over higher-risk clients.
36% of the overdraft were approved beyond the 15 day processing TAT	● Establish real-time dashboards to track processing timelines by analyst and stage, enabling proactive management of delays and accountability. ● Implement workflow automation for document verification, checklist tracking, and approvals to reduce manual bottlenecks and speed up processing. ● Create a fast-track approval process for low-risk customers (e.g., high credit scores, existing clients) to reduce processing time and improve client experience.
0.6% of the collateral are yet to be legally perfected	● Develop a centralized database to monitor collateral documentation status, expiry dates, and perfection stages across the portfolio. ● Set up alerts for pending or overdue collateral perfection to ensure timely follow-up and escalation before risk exposure increases. ● Perform quarterly audits on collateral records to identify gaps and ensure all securities are legally enforceable and properly documented.